Applying AI to digital archives: trust, collaboration and shared professional ethics

将人工智能应用于数字档案：信任、协作与共同的职业伦理

Lise Jaillant; Arran Rees

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Abstract

摘要

Policy makers produce digital records on a daily basis. A selection of records is then preserved in archival repositories. However, getting access to these archival materials is extremely complicated for many reasons—including data protection, sensitivity, national security, and copyright. Artificial Intelligence (AI) can be applied to archives to make them more accessible, but it is still at an experimental stage. While skills gaps contribute to keeping archives 'dark', it is also essential to examine issues of mistrust and miscommunication. This article argues that although civil servants, archivists, and academics have similar professional principles articulated through professional codes of ethics, these are not often communicated to each other. This lack of communication leads to feelings of mistrust between stakeholders. Mistrust of technology also contributes to the barriers to effective implementation of AI tools. Therefore, we propose that surfacing the shared professional ethics between stakeholders can contribute to deeper collaborations between humans. In turn, these collaborations can lead to the building of trust in AI systems and tools. The research is informed by semi-structured interviews with thirty government professionals, archivists, historians, digital humanists, and computer scientists. Previous research has largely focused on preservation of digital records, rather than access to these records, and on archivists rather than records creators such as government professionals. This article is the first to examine the application of AI to digital archives as an issue that requires trust and collaboration across the entire archival circle (from record creators to archivists, and from archivists to users).

政策制定者每天都会产生数字记录，其中一部分随后被保存进档案机构。然而，由于数据保护、敏感性、国家安全和版权等多重原因，获取这些档案材料十分复杂。人工智能（AI）可以被应用于档案，以提高其可获取性，但这种应用仍处于实验阶段。技能缺口固然会使档案继续处于“黑暗”状态，但同样需要关注不信任和沟通不足的问题。本文认为，公务员、档案工作者和学者虽然都拥有由各自职业伦理准则所表达的相似职业原则，但这些原则彼此之间往往没有得到充分沟通。缺乏沟通会导致利益相关者之间产生不信任，而对技术的不信任也构成有效实施 AI 工具的障碍。因此，本文提出，将各方共同拥有的职业伦理明确呈现出来，可以促进人与人之间更深入的协作；这种协作反过来又能够帮助建立对 AI 系统和工具的信任。本研究基于对 30 名政府工作人员、档案工作者、历史学家、数字人文学者和计算机科学家的半结构式访谈。既有研究主要关注数字记录的保存，而非这些记录的开放利用，并且更多聚焦档案工作者，而较少关注政府工作人员等记录创建者。本文首次从整个档案工作链条——从记录创建者到档案工作者，再从档案工作者到使用者——所需要的信任与协作出发，考察 AI 在数字档案中的应用。

For there to be betrayal, there would have to have been trust first. — Suzanne Collins, The Hunger Games

“若要发生背叛，首先必须曾经存在信任。”——苏珊·柯林斯，《饥饿游戏》

1 Introduction

1 引言

Born-digital archives1 have been acquired and preserved for the best part of thirty years. Yet, access to these records remains a complex challenge for the institutions that hold and manage them, as well as the researchers seeking to use them.

Obstacles to access are numerous and include issues with data protection, sensitivity, and copyright. In the case of government archival collections, releasing potentially sensitive and private information could threaten national security and embarrass foreign partners. However, not all collections present the same level of risk, and giving access to digital records is essential to make government accountable and enable the writing of history.

过去近三十年来，档案机构一直在接收和保存原生数字档案 1。然而，无论对保管和管理这些记录的机构，还是对希望使用这些记录的研究者而言，如何获得访问权限仍然是一个复杂挑战。障碍很多，包括数据保护、敏感性和版权问题。对于政府档案而言，公开可能包含敏感或私人信息的材料，可能威胁国家安全，也可能使外国合作伙伴陷入尴尬。不过，并非所有馆藏都具有同等程度的风险；开放数字记录对于实现政府问责和历史书写都十分重要。

As born-digital records increasingly represent the largest part of new accessions, finding solutions to these obstacles is becoming a key priority. Unlocking vast amounts of digital data is not a task that can be done manually. Artificial Intelligence2 (AI) has the potential to make born-digital archives more accessible and usable. Sensitive information can be automatically identified, making possible the release of non-sensitive data. AI can also be used when keyword search is not effective (for example in the case of web archives, which include terabytes of information). An AI-driven process of recommendation—similar to the functionality 'customers who bought this item also bought'—could enable discovery of previously inaccessible archival materials.

随着原生数字记录在新增档案中所占比例不断扩大，解决这些访问障碍已成为优先事项。海量数字数据不可能完全依靠人工来解锁。人工智能 2（AI）有潜力提高原生数字档案的可获取性和可使用性。例如，系统可以自动识别敏感信息，从而使非敏感数据得以开放；当关键词检索效果不佳时，AI 也可以发挥作用，例如面对包含 TB 级信息的网络档案。类似“购买此商品的顾客还购买了”的 AI 推荐机制，也可能帮助研究者发现过去难以接触到的档案材料。

Several projects have recently sought to identify key challenges and find solutions to the problem of locked born-digital archives, using AI and other advanced technologies.3 Yet, AI applied to archives remains at the experimental stage; rarely implemented beyond small data sets and collections identified for experimentation. Expanding the application of AI to vast amounts of archival materials is complicated for many reasons—including the difficulty to recruit specialists with the right skills, and the relatively low salaries in the archive sector. These obstacles could be overcome with additional funding to attract talent, as well as training opportunities for staff who already work in the sector. A thornier problem is the frequent lack of trust and shared professional ethics across the archival circle (from record creators to archivists, and from archivists to researchers).

近年来，一些项目尝试识别关键挑战，并利用 AI 和其他先进技术解决被封闭的原生数字档案问题。3 然而，AI 在档案中的应用仍处于实验阶段，通常只在小型数据集或专门挑选的馆藏中开展试验。若要把 AI 扩展到海量档案材料，还面临许多困难，包括难以招聘具备合适技能的专业人员，以及档案行业相对较低的薪资水平。增加经费以吸引人才、为现有从业人员提供培训，可以缓解这些问题。但更棘手的障碍，是整个档案工作链条中——从记录创建者到档案工作者，再到研究者——经常缺乏信任和共同的职业伦理。

The Oxford English Dictionary defines 'trust' as the 'firm belief in the reliability, truth, or ability of someone or something' and the 'confidence or faith in a person or thing, or in an attribute of a person or thing.' The word was inherited from Germanic languages, and is often associated with protection, shelter, and safety. We trust if we feel safe. On the opposite, we lack trust in a situation that seems risky and unsafe. Building trust is a requirement for many professionals, who need to show that their work is beneficial, or at least innocuous. This is why academics must obtain approval from ethics committees to conduct certain kinds of research (including research with human participants). For the OED, 'ethics' is defined as 'the codes of conduct or moral principles recognized in a particular profession' and other aspects of life. Professionals do not always agree on what is or is not ethical. In the case of new technologies such as AI, the lack of trust and shared professional ethics can lead to a deadlock. For example, archivists will hesitate to release data for computational research if they do not trust that academic researchers will make good use of these data. In this context, AI can do nothing to make digital archives more accessible: the first step is to address the lack of trust and common codes of conduct. Once the trust between humans has been restored, AI can come in as the next step to unlock digital archives.

《牛津英语词典》将“信任”定义为对某人或某事物的可靠性、真实性或能力具有“坚定的相信”，以及对人或事物、或其某种属性具有“信心或信念”。这个词源自日耳曼语系，常与保护、庇护和安全联系在一起。人们在

感到安全时会信任，而在觉得存在风险和不安全时则缺乏信任。许多职业都必须建立信任，因为从业者需要证明其工作是有益的，至少是无害的。这也是学者在开展某些研究——包括涉及人类参与者的研究——之前必须获得伦理委员会批准的原因。OED 将“伦理”定义为“某一职业所认可的行为规范或道德原则”等。专业人士并不总是对什么属于伦理、什么不属于伦理达成一致。对于 AI 这样的新技术，缺乏信任和共同职业伦理可能造成僵局。例如，如果档案工作者不相信学术研究者会妥善使用数据，就会犹豫是否为计算研究开放数据。在这种情况下，AI 本身无法提高数字档案的可获取性：第一步必须先解决人与人之间缺乏信任和共同行为规范的问题；只有在恢复人际信任之后，AI 才可能成为解锁数字档案的下一步。

In this article, we investigate these issues of trust and shared professional ethics using qualitative data from the project ‘Unlocking our Digital Past’ which brought together our team of Digital Humanists with government professionals and archivists. The project was organized in collaboration with the UK Cabinet Office, and although its main focus was Britain, it also involved professionals from North America and Continental Europe. We sought to better understand the barriers to implementing AI-driven tools to make digital archival collections more accessible, and identify key concerns from stakeholders in the entire archival process; from the civil servants and policy makers creating the documents and records that end up in The National Archives, to the archivists and GLAM (Galleries, Libraries, Archives and Museums) professionals preserving the digital collections, and the academic researchers attempting to access and use them.

本文利用“解锁我们的数字过去”（Unlocking our Digital Past）项目的质性数据，研究信任与共同职业伦理问题。该项目将我们的数字人文团队与政府工作人员和档案工作者联系起来，并与英国政府内阁办公室合作开展。虽然项目重点在英国，但也有来自北美和欧洲大陆的专业人士参与。我们的目标是更好理解实施 AI 工具、提高数字档案馆藏可获取性时面临的障碍，并识别整个档案流程中各类利益相关者的主要关切：从创建最终进入英国国家档案馆的文件和记录的公务员、政策制定者，到保存这些数字馆藏的档案工作者和 GLAM（画廊、图书馆、档案馆和博物馆）专业人士，再到试图获取和使用这些材料的学术研究者。

The research is informed by semi-structured interviews with 30 government professionals, archivists, historians, digital humanists, and computer scientists, and was driven by two key research questions: What are the potential applications of AI to make borndigital archives more accessible and usable? And what are the barriers to implementing them? The project found that many of the concerns about the application of AI are shared between stakeholders in the civil service, GLAM sector, and academia. Yet, collaborative conversations, events and projects between the three groups are rare.

本研究依据对 30 名政府工作人员、档案工作者、历史学家、数字人文学者和计算机科学家的半结构式访谈，并围绕两个核心问题展开：AI 有哪些潜在应用能够使原生数字档案更易获取和使用？实施这些应用又面临哪些障碍？项目发现，公务员体系、GLAM 领域和学术界的利益相关者对 AI 应用存在许多共同担忧，但这三类群体之间的协作性对话、活动和项目却十分少见。

This article argues that although civil servants, GLAM professionals and academics have similar professional principles articulated through professional codes of ethics, these are not often communicated to each other. This lack of communication leads to feelings of mistrust between stakeholders (for example, when archivists do not trust that researchers will make the right decision when confronted with sensitive information). Mistrust of technology also contributes to the barriers to effective implementation of AI tools. Therefore, we propose that surfacing the shared professional ethics between stakeholders can contribute to deeper collaborations between humans. In turn, these collaborations can lead to the building of trust in AI systems and tools. If record creators and archivists trust that researchers will treat data in an ethical way, they will be more eager to share these data for research (including computational research). Trust in other humans can lead to trust in technology.

本文认为，公务员、GLAM 专业人士和学者虽然都有通过职业伦理准则表达的相似职业原则，但这些原则很少被彼此沟通。这种沟通不足会造成利益相关者之间的不信任，例如档案工作者可能不相信研究者在遇到敏感信息时能够作出正确决定。对技术的不信任也妨碍 AI 工具的有效实施。因此，本文提出，明确呈现不同利益相关者共享的职业伦理，可以促进人与人之间更深入的协作，而这些协作又能够建立对 AI 系统和工具的信任。如果记录创建者和档案工作者相信研究者会以合乎伦理的方式处理数据，他们就更愿意分享数据用于研究，包括计算研究。对人的信任可以进一步转化为对技术的信任。

The three sections of this article develop the argument that shared professional ethics can contribute to closer collaborations between record creators, archivists and researchers, leading to the building of trust in AI tools. The first section looks at the general context and related work on the key barriers to accessing borndigital collections, and the potential applications of AI that could unlock these collections. It also explains our methodology and approach. Drawing on our series of interviews, the second section then focuses on the issues of mistrust of other stakeholders and mistrust of technology, which contribute to the lack of access to digital collections. The third section explores our recommendation to build up trust through collaboration and shared professional ethics, in order to step up the application of AI and new technologies to digital archives. Unquestioning trust in AI is neither desirable nor possible. Yet, the building of trust in collectively developed and explainable AI tools across the civil service, GLAM sector and academia will help make digital archives more accessible and usable.

本文的三个主要部分进一步论证：共同的职业伦理可以推动记录创建者、档案工作者和研究者之间更紧密的协作，并由此建立对 AI 工具的信任。第一部分介绍原生数字馆藏难以开放的总体背景和相关研究，以及 AI 可能如何解锁这些馆藏，同时说明本文的方法与研究路径。第二部分基于访谈，讨论对其他利益相关者和对技术的不信任如何限制数字馆藏的开放。第三部分则提出，通过协作和共同职业伦理建立信任，以扩大 AI 和新技术在数字档案中的应用。对 AI 毫无保留的信任既不可取，也不可能；但公务员体系、GLAM 领域和学术界如果能够共同开发可解释的 AI 工具并建立合理信任，将有助于提高数字档案的可获取性和可使用性。

2 Context, related work, and methodology

2 背景、相关研究与方法

Substantial work has been undertaken over the past thirty years or so to establish effective practices for preserving born-digital materials in the GLAM sector (Deegan and Tanner, 2006; Delve and Anderson, 2014; Corrado and Sandy, 2017; Harvey and Weatherburn, 2018; Owens, 2018). Influential support organizations like the Digital Preservation Coalition have helped embed and sustain the work of digital preservation practitioners across a range of cultural heritage organizations. However, while preservation processes are becoming ever more integrated into the core work of GLAM institutions, processes for providing access to those same born-digital collections has been slower to develop (Jaillant, 2019). There are multiple reasons for this, from content-related legal issues such as privacy concerns and copyright, to technology issues that make it difficult to provide access to materials in appropriate formats.

过去约三十年间，GLAM 领域已经开展了大量工作，以建立保存原生数字材料的有效实践。数字保存联盟（Digital Preservation Coalition）等重要支持机构，帮助数字保存工作者在各类文化遗产机构中建立并维持相关实践。然而，虽然数字保存流程越来越成为 GLAM 机构的核心工作，为同一批原生数字馆藏提供访问的机制却发展得更慢。原因很多，既有隐私、版权等内容层面的法律问题，也有技术问题，使机构难以用合适的格式提供材料。

The legal concerns mostly relate to the General Data Protection Regulations (GDPR) in the European Union, the Data Protection Act 2018 in the UK and Copyright legislation. The GDPR and Data Protection legislation place a responsibility to build in ‘privacy by design’ on organizations and institutions handling personal data, but also provide exemptions to aspects of the regulations when personal data is being archived in the public interest (Lomas, 2019). Whilst The National Archive’s Guide to Archiving Personal Data (2018) makes clear that the updated data protection does not prevent archiving, it does influence how archival institutions need to articulate the purposes of archiving personal data and how they provide access to collections containing it (The National Archives, 2018). Records deemed to contain personal data are carefully managed and normally kept closed from public access during the lifetime of the individual the data is relating to (The National Archives suggests that a lifespan of 100 years should be presumed, unless proven otherwise).4 However, in some circumstances access to these records can be granted, and decisions need to be taken with respect to the safeguarding measures defined in the Data Protection legislation, and fully documented by the archive (The National Archives, 2018).

法律方面的担忧主要涉及欧盟《通用数据保护条例》（GDPR）、英国《2018 年数据保护法》以及版权法律。GDPR 和数据保护立法要求处理个人数据的组织和机构实行“隐私保护设计”，但当个人数据出于公共利益被归档时，也允许适用部分豁免。英国国家档案馆的《个人数据归档指南》（2018）明确指出，更新后的数据保护规定并不阻止归档，但确实会影响档案机构如何说明保存个人数据的目的，以及如何开放包含此类数据的馆藏。被认为含有个人数据的记录通常会受到严格管理，并在相关个人生存期间对公众关闭（国家档案馆建议，在没有相

反证据时假定寿命为 100 年）。4 不过，在某些情况下仍可开放这些记录，但必须根据数据保护法律规定的保障措施作出决定，并由档案机构完整记录相关过程。

Additional concerns relate to Copyright legislation which has an impact on how archival material can be processed and made accessible. The application of copyright law on born-digital materials in GLAM institutions is an ongoing and complex issue. Many borndigital collections contain content from multiple copyright holders, described at varying levels of detail and often without proof of consent. For example, an archived website may have personal information, images, creative writing, or videos embedded within it. It is not possible to assume that all the content belongs to the person or organisation that created it, and it is often not possible to trace who the copyright holder is. Many GLAM institutions with web archives face difficult decisions about the amount of their collections they can make available whilst managing the potential risks of making it available (Hockx-Yu, 2014; Vlassenroot et al., 2019). Copyright legislation in the UK and in the EU has acknowledged the digitization of physical material as a form of preservation in GLAM institutions and has developed exemptions to support this. However, the management of born-digital material is largely ignored in the legal context. Creating preservation and access copies of born-digital content remains a grey area (Koscık and Myska, 2019).

另一类担忧来自版权法，它会影响档案材料如何被处理和开放。如何将版权法适用于 GLAM 机构中的原生数字材料，是一个持续存在且十分复杂的问题。许多原生数字馆藏包含多个版权持有者的内容，描述详略不一，且往往缺少授权证明。例如，一个被归档的网站可能嵌入个人信息、图片、创作性文字或视频，不能假定所有内容都属于创建该网站的个人或组织，也常常无法追踪实际版权持有人。许多拥有网络档案的 GLAM 机构因此必须在开放多少馆藏和控制开放风险之间作出艰难选择。英国和欧盟的版权法已承认把实体材料数字化作为 GLAM 机构的一种保存方式，并为此制定豁免规定；但在法律层面，原生数字材料的管理仍在很大程度上被忽视，为原生数字内容制作保存副本和访问副本仍属于灰色地带。

With so many digital collections closed to users due to these issues with data protection and copyright, AI has been presented as a possible solution to the problem of locked archives (Jaillant and Caputo, 2022; Jaillant 2022a, 2022b). For the purposes of this article, we refer to AI in a broad sense. Instead of focusing on particular subsets of AI such as machine learning, computer vision, or natural language processing, we consider AI as a broad suit of computational approaches used to make decisions and complete tasks. Although often discussed through the lens of new and emerging technologies, AI has a longer history, with waves of excitement and periods of disinterest (Jackson, 2019; Agar, 2020). The current period is characterized by sustained development, and AI technologies are increasingly featuring in all aspects of our everyday lives. The role of AI technologies in making born-digital archives more accessible and usable is increasingly acknowledged as an important direction for record creators and managers (including civil servants responsible for knowledge, information, and records management), GLAM professionals, and researchers interested in using the archival material.

由于数据保护和版权问题导致大量数字馆藏对用户关闭，AI 被提出作为解决“封闭档案”的可能方案。本文对 AI 采用宽泛定义，不局限于机器学习、计算机视觉或自然语言处理等特定子领域，而是将其视为一组用于作出判断和完成任务的计算方法。虽然 AI 经常被作为新兴技术来讨论，但它有更长的历史，经历过多轮热潮和冷却。目前则处于持续发展时期，AI 技术日益进入日常生活的各个方面。利用 AI 提高原生数字档案的可获取性和可使用性，也越来越被记录创建者和管理者、GLAM 专业人士以及希望利用档案材料的研究者视为重要方向。

Examples of AI tools used to increase the accessibility and usability of digital archives include: the automatic sensitivity classification of government records before being released through Freedom of Information requests or transferred to national archives (The National Archives UK, 2016; Mcdonald et al., 2020a; Baron et al., 2020); experimentations with computer vision to enhance the discovery of visual archival collections (Angelova et al., 2020); and algorithmic search tools (Nix and Decker, 2021). Additionally, scholarship and research networks, such as AURA, AEOLIAN and AI4LAM bring together multiple stakeholders interested in the application of AI to archival collections.

用于提高数字档案可获取性和可使用性的 AI 应用包括：在政府记录通过信息自由申请公开或转交国家档案机构之前，自动进行敏感性分类；利用计算机视觉增强视觉档案馆藏的发现；以及使用算法检索工具。此外，AURA、AEOLIAN 和 AI4LAM 等学术和研究网络也把多个关注 AI 档案应用的利益相关者聚集起来。

As awareness of AI technologies increase, so does public scrutiny, and ethical questions, including ones on transparency and bias, begin entering public discourses (Fast and Horvitz, 2017; Cave et al., 2019; Kelley et al., 2021). Calls to develop more widely agreed standards and processes of enforcement for the algorithms that underpin a lot of AI technologies

continue to grow. Projects such as Assembling Accountability have sought to develop frameworks to support the assessment of algorithmic impacts when utilising AI-drive tools. This project argues that impact assessments are 'a promising model of algorithmic governance because it bundles an account of potential and actual harms of a system with a means for identifying who is responsible for their remedy' (Moss et al., 2021, p. 1).

随着公众对 AI 技术的了解增加，审视也随之增强，透明度和偏见等伦理问题开始进入公共讨论。人们越来越呼吁为支撑 AI 技术的算法建立更广泛认可的标准和执行机制。"组装问责"（Assembling Accountability）等项目尝试建立框架，用于评估使用 AI 工具时算法产生的影响。该项目认为，影响评估是一种很有潜力的算法治理模式，因为它既记录系统可能和实际造成的伤害，也提供识别应由谁负责补救的方法。

Civil servants, GLAM sector professionals and academic researchers increasingly engage with ethical issues related to AI systems. In 2018, the UK government established the Centre for Data Ethics and Innovation to provide independent advice on measures needed to ensure safe, ethical, and innovative uses of AI. One of its core missions is to help public sector partners to use data and AI in a way that commands and retains public trust.5 And in October 2020, the UK Parliament's Office of Science and Technology released a note on interpretable machine learning, to ensure that ML systems are designed and deployed in an ethical and responsible way (Christie, 2020). In the GLAM sector, an emerging body of work addresses transparency, accountability, and bias in libraries and archives (Padilla, 2019; Cordell 2020), as well as museums (Murphy and Villaespesa, 2020). This work explores the public discourse around ethical uses of AI. However, it is not sufficiently connected to considerations of professional ethics across the civil service, GLAM sector, and academic researchers.

公务员、GLAM 专业人士和学术研究者越来越多地参与 AI 系统相关的伦理问题。2018 年，英国政府成立数据伦理与创新中心，为安全、合乎伦理且具有创新性的 AI 使用提供独立建议，其核心使命之一，就是帮助公共部门以能够获得并维持公众信任的方式使用数据和 AI。2020 年 10 月，英国议会科学与技术办公室又发布关于可解释机器学习的说明，以确保机器学习系统以合乎伦理和负责任的方式设计与部署。GLAM 领域也逐渐形成关于图书馆、档案馆和博物馆中透明度、问责和偏见问题的研究。不过，这些讨论与公务员体系、GLAM 领域和学术研究者之间的职业伦理联系仍不充分。

Ethical questions in relation to new technologies begin to intersect with the codes of ethics that govern professional practice in those sectors. Civil servants, GLAM professionals, and academics all subscribe to some form of ethical practice, whether it be through established codes of practice, like the Civil Service Code6 and the International Council on Archives' Code of Ethics,7 or processes of ethical approval seen in university-based academic research. As well as the ethical practices embedded within the civil service, GLAM sector and academic professions, the application of AI technologies is also subject to ethical codes and guidelines, although less established and not formally agreed amongst private companies, research institutions, and public sector bodies (Jobin et al., 2019; Mittelstadt, 2019; Hagendorff, 2020). One study found that there were eighty-four sets of ethical principles or guidelines for AI published globally (Jobin et al., 2019). These existing frameworks lack specific actions and coordination across professional sectors.

与新技术相关的伦理问题，开始与这些行业中规范职业实践的伦理准则相交。公务员、GLAM 专业人士和学者都遵循某种形式的伦理实践，例如《公务员守则》、国际档案理事会《伦理守则》，或大学学术研究中的伦理审批程序。除这些职业本身已有的伦理实践外，AI 技术的应用也受到伦理规范和指南约束，只是这些规则尚不成熟，在私营企业、研究机构和公共部门之间也尚未正式统一。一项研究发现，全球已发布 84 套 AI 伦理原则或指南，但现有框架缺乏具体行动方案以及跨专业领域的协调。

Although there is no single agreed set of ethical principles in relation to AI, the notions of transparency, justice, and fairness, non-maleficence, responsibility, and privacy are the most commonly discussed (Jobin et al., 2019, p. 391).8 These five principles can also be found in the ethical codes and professional guidelines followed by our interviewees. Evaluating the effectiveness of these ethical codes and guidelines is outside the scope of this article. However, as Iason Gabriel notes, we should be looking 'more closely at principles that would be supported by a global overlapping consensus of opinion' to support the use of AI (Gabriel, 2020, p. 433). Working through shared principles, expressed in codes of ethics, presents itself as a productive place to start building trust in the application of AI to borndigital archives.

虽然关于 AI 尚不存在一套统一认可的伦理原则，但最常被讨论的包括透明度、正义与公平、不伤害、责任和隐私。这五项原则同样出现在受访者所遵循的伦理准则和职业指南中。本文不评价这些伦理准则和指南本身是否有

效，但正如 Iason Gabriel 指出的，在支持 AI 使用时，应更仔细寻找能够得到全球“重叠共识”支持的原则。以伦理准则中已经存在的共同原则为基础，是为 AI 应用于原生数字档案建立信任的一个有效起点。

The common ethical notions of transparency, justice and fairness, non-maleficence, responsibility, and privacy noted by Jobin et al. do not inherently conflict with the principles expressed in the professional codes and practices seen in the civil service, GLAM sector, and academic researchers. The Civil Service Code specifies integrity, honesty, objectivity, and impartiality as the key values for civil servants, and these principles permeate into every facet of how civil servants approach their professional roles. A number of these principles are underpinned through laws and regulations like the Freedom of Information Act and Public Records Act that build in a culture of accountability into the documents that civil servants and public office holders create every day.

Jobin 等人所概括的透明度、正义与公平、不伤害、责任和隐私，并不与公务员、GLAM 专业人士和学术研究者的职业规范相冲突。《公务员守则》将廉正、诚实、客观和公正列为核心价值，这些原则渗透于公务员职业角色的各个方面，并由《信息自由法》《公共记录法》等法律和制度加以支撑，从而在公务员和公职人员日常创建的文件中建立问责文化。

Moreover, codes of ethics and practice have also been instrumental in the professionalisation of the GLAM sector. In 1986, the International Council of Museums (ICOM) released their code of ethics.9 Ten years later, the International Council on Archives (ICA) introduced their own ethical framework. The International Federation of Library Associations (IFLA) approved a code in 2012.10 These codes are often supplemented by country specific codes, and the United Kingdom also has codes of ethics from the UK Museums Association and CILIP (the UK’s library and information association).11 Integrity, impartiality, objectivity, privacy, and co-operative working feature as principles for archivists to work with.

伦理与实践准则也推动了 GLAM 领域的专业化。1986 年，国际博物馆协会（ICOM）发布伦理守则；十年后，国际档案理事会（ICA）推出自己的伦理框架；国际图书馆协会与机构联合会（IFLA）则在 2012 年通过相关准则。这些国际规范通常还会由各国自己的行业准则加以补充。廉正、公正、客观、隐私和合作工作等原则，都成为档案工作者应遵循的职业规范。

As well as the codes that govern civil servants and GLAM sector professionals, academic researchers also work within ethical frameworks. Ethics within academic research is governed by a set code that is integrated into individual university strategies and research approval processes. The UK Research Integrity Office (UKROI) is thus responsible for the UK’s Code of Practice for Researchers.12 Loughborough University, home to the ‘Unlocking our Digital Past’ project, has an Ethical Policy Framework informed by the UKRIO code of practice, designed to build ethical considerations into research practices. This article does not claim that all professionals working in the civil service, GLAM sector and academia consult their codes of practice or ethics before they make new decisions or undertake a new task, but that these codes broadly influence the principles and areas of concern for the sectors.

除公务员和 GLAM 专业人士的行业准则外，学术研究者同样在伦理框架中开展工作。学术研究伦理通常通过统一规范，嵌入各大学的研究战略和审批流程。英国研究诚信办公室（UKRIO）负责英国《研究人员实践守则》；“解锁我们的数字过去”项目所在的拉夫堡大学，也依据这一守则建立伦理政策框架，将伦理考量纳入研究实践。本文并不是说所有公务员、GLAM 专业人士和学者在每次作出决定或开展新任务前都会查阅这些准则，而是认为这些准则总体上塑造了各行业的原则和关注重点。

Concepts of trust, transparency, and professional ethics emerged as a recurring theme throughout ‘Unlocking our Digital Past’. The project was part of an ongoing collaborative relationship between Loughborough University and the Cabinet Office of the UK Government that focuses on the application of AI to born-digital archives. The project facilitated a series of workshops aimed at increasing interdisciplinary dialogue between civil servants, GLAM sector professionals, and academics. After obtaining approval from the Ethics Committee at our institution, our team of Digital Humanists conducted thirty interviews with key thinkers and practitioners who have a stake in the accessibility and usability of born-digital archives now and in the future. Our background is in cultural history, literary studies and museum studies, rather than computer science. This has informed the nature of our questions: instead of focusing on the technical aspects of AI, the questions we asked centred on the key obstacles to making born-digital and digitised collections more accessible, and on the possible solutions to these issues. We were interested in the day-to-day practice of professionals—from record creators who need to prepare archival records ahead of transfer to The National Archives and other repositories, to archivists who need to preserve and make accessible these records. We also interviewed historians who need access to archives (including born-

digital archives) for their research. A common thread of these interviews was the lack of trust in other stakeholders and in technology. In turn, this mistrust has an impact on the accessibility and usability of digital archives.

信任、透明度和职业伦理，是“解锁我们的数字过去”项目中反复出现的主题。该项目建立在拉夫堡大学与英国政府内阁办公室持续合作的基础上，重点关注 AI 在原生数字档案中的应用。项目通过一系列工作坊促进公务员、GLAM 专业人士与学者之间的跨学科对话。在获得机构伦理委员会批准后，数字人文团队访谈了 30 位与当前和未来原生数字档案可获取性、可使用性密切相关的重要研究者和实践者。团队成员的背景是文化史、文学研究和博物馆研究，而非计算机科学，这也影响了提问方式：我们没有把重点放在 AI 技术细节上，而是关注原生数字和数字化馆藏开放时最重要的障碍及其可能解决方案。受访者涵盖需要在移交国家档案馆前整理档案的记录创建者、保存和开放记录的档案工作者，以及需要使用档案——包括原生数字档案——开展研究的历史学家。访谈中贯穿始终的一条线索，就是对其他利益相关者和技术的不信任，而这种不信任又直接影响数字档案的可获取性和可使用性。

3 Mistrust of other stakeholders and of technology

3 对其他利益相关者与技术的不信任

Collaborative discussions between professionals in government, GLAMs and academia are the exception rather than the norm. This, in turn, has an impact on the way professionals see other professionals, often leading to misunderstandings. James Baker, who studied history and worked as a curator at the British Library before moving to academia, gave an example of this lack of dialogue between archivists and researchers. While archivists see appraisal and selection of records as a central aspect of their role, researchers often expect complete archives. In other words, academics view archivists as record keepers, rather than professionals who must make a selection and throw away records that lack lasting value. This task of appraisal is particularly complicated due to the scale of born-digital archives. Baker said:

政府、GLAM 机构和学术界专业人士之间的协作性讨论并非常态，而是例外。这会影响不同专业群体彼此如何看待对方，并经常造成误解。James Baker 曾学习历史、在大英图书馆担任馆员，后来进入学术界。他以档案工作者与研究者之间缺乏对话为例：档案工作者把鉴定和筛选记录视为核心职责，而研究者却常常期待得到“完整档案”。换言之，学者容易把档案工作者想象成单纯的记录保管者，而不是必须筛选并淘汰缺乏长期价值材料的专业人士。在原生数字档案规模不断扩大的情况下，这种鉴定工作尤其复杂。Baker 说：

> *I always feel very sorry for my archivist friends every time historians turn round and say, well, you must collect everything, and why aren't you trying, you know, we need all this stuff and it's going to be a digital dark age. And they're like, it's not going to be a digital dark age, we always do selection, you know, it's just that selection issues are kind of way more complicated now in the age of the born digital.13*
>
> *“每次历史学家转过头说，你们必须把所有东西都保存下来，为什么不努力做到这一点，我们需要这些材料，否则就会进入数字黑暗时代时，我都很同情我的档案工作者朋友。而他们会说，这不会成为数字黑暗时代，我们一直都在做筛选，只是在原生数字时代，筛选问题变得复杂得多。”13*

There was a consensus among our interviewees who engage closely with born-digital records: keeping and describing everything is not a viable option. Clifford Lynch, Executive Director of the Coalition for Networked Information, pointed out that in the very early days of the internet, there were attempts to apply library-style cataloguing to internet resources. But these ‘were rapidly abandoned because the scale was just unmanageable’.14 The Google-style, computational approach to records became the norm to search records at scale.

密切接触原生数字记录的受访者普遍认为，保存并描述所有内容并不可行。网络信息联盟执行主任 Clifford Lynch 指出，互联网早期曾有人尝试用图书馆式编目方法处理网络资源，但由于规模完全无法管理，这些尝试很快被放弃。此后，类似 Google 的计算式检索方法成为大规模搜索记录的常态。

The problem of scale is particularly acute for government digital records. Jason Baron, who served for 13 years as Director of Litigation for the US National Archives and Records Administration (NARA) before moving to the private sector and then academia, gives the example of US presidents’ email records:

规模问题在政府数字记录中尤其突出。Jason Baron 曾在美国国家档案与记录管理局（NARA）担任诉讼主任 13 年，之后进入私营部门和学术界。他以美国总统电子邮件为例：

> *The National Archives in the US today has 500 million emails from the Regan administration to the Obama administration. The Trump emails haven't been counted yet or haven't been fully processed. But it has 500 million distinct emails, and then more than a billion pages because those emails have attachments.15*
>
> *"美国国家档案馆目前保存着从里根政府到奥巴马政府的 5 亿封电子邮件。特朗普政府的邮件还没有统计，或尚未全部处理。但已经有 5 亿封独立邮件，加上附件之后，总页数超过 10 亿页。"15*

In this role at NARA, Baron pushed for an approach that would diminish the number of government emails transferred to archival repositories. This CAPSTONE approach adopted in 2013 led to an email preservation policy based on the seniority of the record creator.16 Individuals at the top or near the top of an agency have all their emails permanently preserved, while others see their emails destroyed after a certain period (generally seven years).

在 NARA 任职期间，Baron 推动减少转移至档案机构的政府电子邮件数量。2013 年采用的 CAPSTONE 方案，按照记录创建者的职位高低制定电子邮件保存政策：机构最高层或接近最高层人员的全部邮件永久保存，其他人的邮件则在一定年限后销毁，通常是七年。

CAPSTONE addressed the issue of email exponential increase, but emails are of course not the only born-digital records produced by US government. Leslie Johnston (Director of Digital Preservation at NARA) notes that the Trump administration produced about 500 terabytes of archival records—compared to 200 terabytes for the two terms of the Obama administration.17 Libraries and archival institutions lack the staff necessary to handle enormous amounts of data. Baron gives the example of the Clinton Presidential Library, which has six archivists and 'a 10 million document queue estimate for Freedom of Information Act requests'.18 This leads to a lack of accountability in the short term, and risk impacting the cultural memory in the long term.

CAPSTONE 缓解了电子邮件数量指数增长的问题，但政府产生的原生数字记录当然不只有电子邮件。NARA 数字保存主任 Leslie Johnston 指出，特朗普政府产生了大约 500TB 的档案记录，而奥巴马政府两个任期合计约为 200TB。图书馆和档案机构缺少足够人员处理如此庞大的数据。Baron 举例说，克林顿总统图书馆只有六名档案工作者，却面对大约一千万份等待处理的信息自由法申请文件。这在短期内造成问责不足，在长期则可能影响文化记忆。

Jason Baron and others have argued that automation is a necessity rather than a choice. 'How do you search a billion objects?', asks Baron. 'Well, you don't do it manually, as an archivist, and you don't do it with keyword searches, although you can try, but those are terribly inefficient'.19 Indeed, an approach based on keyword search does not work well with collections characterized by their huge size and lack of metadata. For instance, a search for 'Brexit' on the UK web archive gives 63,968,885 results.20 As Leontien Talboom (Doctoral Researcher at The National Archives UK and University College London) told us, analysing each record manually is not an option.21 Instead, a computational approach is the only way to make sense of this mass of results. 'How can they have any success of opening a substantial amount of history for the 21st century and before, if they don't use machine learning techniques to perform searches?' Baron inquires.22 It should be noted that AI-assisted searches are not perfect solutions, since they might also miss relevant results.23

Jason Baron 等人认为，自动化已经不是选择，而是必需。"十亿个对象要怎么搜索？"Baron 问道。"你不可能作为档案工作者手工做，也不能只依靠关键词搜索；当然你可以试，但效率非常低。"对于规模巨大且缺少元数据的馆藏，关键词检索确实效果不佳。例如，在英国网络档案中搜索"Brexit"会得到 63,968,885 条结果。Leontien Talboom 指出，逐条人工分析根本不可行，只能依靠计算方法理解如此庞大的结果集合。Baron 进一步追问，如果不用机器学习技术进行检索，档案机构又如何可能成功开放大量 21 世纪及更早时期的历史材料？当然，AI 辅助检索也并非完美，它同样可能漏掉相关结果。

The application of AI and machine learning to archival records is not a new thing. In the early 2000s, Jason Baron was part of the government legal team involved in a giant tobacco suit against Philip Morris and other tobacco companies. Key information was scattered in the Clinton email archive, which included 30 million emails. Baron decided to seek out information scientists and computer scientists to figure out a better way for lawyers to search, using other techniques than keyword searching.24 At around the same time, commercial firms started marketing eDiscovery software that drew on

machine learning to search and find information in vast amounts of data. The archive sector on both sides of the Atlantic then investigated the potential of eDiscovery in solving the challenges of born-digital records. In 2016, a report from The National Archives (TNA) in the UK concluded that 'technology-assisted review using eDiscovery software can support government departments during appraisal, selection and sensitivity review as part of a born-digital records transfer' to TNA (The National Archives UK, 2016, p. 5).

把 AI 和机器学习应用于档案记录并非新鲜事。2000 年代初，Jason Baron 作为政府法律团队成员参与针对 Philip Morris 等烟草公司的大型诉讼，关键信息散落在包含 3000 万封邮件的克林顿邮件档案中。他因此寻找信息科学家和计算机科学家，探索关键词检索之外的搜索方法。大约同一时期，商业公司开始推出利用机器学习在海量数据中搜索信息的电子发现（eDiscovery）软件。大西洋两岸的档案领域随后也研究电子发现技术解决原生数字记录难题的潜力。2016 年，英国国家档案馆的一份报告认为，使用电子发现软件进行技术辅助审查，可以支持政府部门在原生数字记录移交过程中开展鉴定、筛选和敏感性审查。

Researchers such as Graham McDonald (Lecturer in Information Retrieval at the University of Glasgow) have shown that AI and machine learning can indeed unlock government archives, using two main approaches: 'protect then search' or 'search then protect'.25 McDonald's work has used mostly the first approach. Tools that his team developed can help humans identify the sensitivities in a particular collection, and then make informed decisions before transferring records to archival institutions. For example, record creators can choose to protect sensitive information by redacting it, ahead of transfer to archives. This process lowers the level of risk in providing access to that collection.

Graham McDonald 等研究者表明，AI 和机器学习确实可能解锁政府档案，主要有两种路径：“先保护、后搜索”或“先搜索、后保护”。McDonald 的研究主要采用第一种：由团队开发的工具帮助人类识别特定馆藏中的敏感内容，再在记录移交档案机构之前作出有依据的决定。例如，记录创建者可以在移交前对敏感信息进行遮蔽处理，从而降低开放该馆藏的风险。

The second approach consists in making a collection entirely accessible, with a search framework that hides sensitive information when it comes across it. The machine is trained using a subset of data identified as sensitive or confidential and is then expected to identify sensitivity in larger datasets. The problem is that there is no universal definition of sensitivity. It largely depends on the context, and the machine is not always able to correctly identify contextual information that makes a record sensitive.26

第二种方法是让整个馆藏都可以访问，但在检索框架遇到敏感信息时自动将其隐藏。机器先使用被标记为敏感或机密的数据子集进行训练，再尝试在更大数据集中识别敏感性。问题在于，“敏感”并没有统一定义，它高度依赖具体语境，而机器并不总能正确识别使某条记录变得敏感的上下文信息。

While Graham McDonald recognizes that sensitivity review is still an imperfect process, the risk of giving access to archival records is worth taking, according to Jason Baron. AI 'will get a tremendous amount of return' even though it comes with a risk of bias.27 The most important thing is to allow researchers, journalists, historians, and other users to make discoveries that will serve the public good. The stakes are lower than in the criminal justice system, where AI has in the past led to disastrous results [e.g. algorithms used in the USA to evaluate the risk of prisoners to re-offend were shown to be racially biased (Wadsworth et al., 2018)]. Despite the risk that comes with AI, the police and justice system on both sides of the Atlantic see advanced technologies as a necessary tool in the fight against crime. In 2014, the UK Home Office released a report on eDiscovery in digital forensic investigations (Lawton et al., 2014). Four years later, it announced the development of new technology to automatically detect terrorist content on online platforms.28

Graham McDonald 承认敏感性审查仍不完善，但 Jason Baron 认为，为开放档案承担一定风险是值得的。即使 AI 存在偏见风险，它仍可能“带来巨大回报”。最重要的是让研究者、记者、历史学家及其他用户获得发现，从而服务公共利益。与刑事司法系统相比，档案领域的风险较低；在刑事司法中，AI 过去曾造成严重后果，例如美国用于评估囚犯再犯风险的算法被证明存在种族偏见。尽管存在风险，大西洋两岸的警务和司法系统仍把先进技术视为打击犯罪的重要工具。

If AI is routinely deployed by police officers and judges to analyse large datasets, why is it rarely applied to government archives? It is still exceptional to use AI for sensitivity review, or to find results in vast digital collections. In order to train AI systems, access to data is needed. And this notion of access (even limited access to training datasets) is problematic for many record creators and archivists. When users are not trusted to make the right decisions, it seems logical to close entire

collections—or to restrict access only to groups who are seen as trustworthy. For example, web archives generally require users to travel to the library due to copyright reasons. Content is available onsite via browsers, rather than offsite via downloading data. Jane Winters, Professor of Digital Humanities at the University of London, notes that Denmark has a researcher exception to this restrictive access policy. Researchers attached to a Danish university, as well as overseas collaborators, can get remote access to web archive data anywhere in the world. 'You have to sign waivers and so on that you will treat the data appropriately', Winters said. This can be seen as a first step 'to start to think about opening things up'.29

如果警察和法官已经经常使用 AI 分析大型数据集，为什么政府档案中却很少使用？目前，利用 AI 进行敏感性审查或在海量数字馆藏中查找结果仍属例外。训练 AI 系统需要数据访问，而访问本身——哪怕只是有限访问训练数据——对许多记录创建者和档案工作者而言都很敏感。当用户不被信任能够作出正确判断时，关闭整套馆藏，或只向被视为可信的群体开放，就显得合乎逻辑。例如，网络档案往往因版权原因要求用户亲自到馆使用，只能现场通过浏览器查看，而不能在馆外下载数据。Jane Winters 指出，丹麦对研究者设有例外政策：与丹麦大学有关联的研究者及海外合作者，只要签署承诺妥善处理数据的文件，就可以在世界任何地方远程访问网络档案。这可以被视为逐步开放的第一步。

The risk of releasing potentially sensitive data should be balanced against the risk of keeping records dark and inaccessible. Indeed, access is at the centre of ethical codes in the GLAM sector. The Museums Association puts 'public engagement and public benefit' as the first principle in its Code of Ethics. More specifically, the code states that museums and other cultural heritage organisations should 'provide public access to, and meaningful engagement with, museums, collections, and information about collections without discrimination'. Likewise, the UK's library and information association CILIP declares that 'preservation and continuity of access to knowledge' is central to libraries' mission. Similarly, the International Council on Archives asserts that 'archivists should promote the widest possible access to archival material and provide an impartial service to all users'. Of course, access is not the only aspect that GLAM professionals need to consider. Codes of ethics include references to privacy and copyright as limits to access. For the ICA, 'archivists should respect both access and privacy'. Yet, the balance is often skewed towards privacy and data protection.

公开潜在敏感数据的风险，应与让记录继续处于“黑暗”和不可访问状态的风险相权衡。事实上，“访问”处于 GLAM 领域伦理准则的核心。英国博物馆协会把“公众参与和公共利益”列为伦理守则的第一原则，并要求博物馆和其他文化遗产机构在不歧视的前提下，为公众提供馆藏及相关信息的访问和有意义的参与。英国图书馆与信息协会 CILIP 也把“保存并持续提供知识访问”视为图书馆使命的核心；国际档案理事会同样指出，档案工作者应促进尽可能广泛的档案访问，并向所有用户提供公正服务。当然，访问并不是唯一考虑，伦理准则也把隐私和版权视为限制。ICA 要求档案工作者“既尊重访问，也尊重隐私”，但现实中的平衡往往更偏向隐私和数据保护。

While archivists must respect legal frameworks and protect the private data of individuals, they could also take limited risks and make some collections available without necessarily checking all materials. The need for trust and shared codes of practice and conduct was highlighted by our interviewees. Jane Winters (University of London) gave the example of personal archives, rather than web archives and other large-scale collections. 'Relationships of trusts between depositing authors, archivists, and researchers' are central. However, these relationships are also 'time-consuming' and 'excluding because if you happen to know the librarians it's much easier to have those conversations than if you're coming to it cold'. In other words, access is often decided on a one-to-one basis based on preliminary discussions with archivists. 'Having those negotiations and building relationships of trust around the use of this material' can unlock previously closed collections.30 Winters's comments echo the experience of literary scholars who have had access to previously closed collections. Lise Jaillant has thus written about her experience of accessing the email archive of the British writer Ian McEwan at the Harry Ransom Center (HRC) in Texas, which is officially inaccessible to researchers. Access was granted through personal contacts, and the fact that the HRC archivist was participating in the same research project (Jaillant 2019).

档案工作者必须遵守法律框架、保护个人私人数据，但也可以承担有限风险，在不逐一检查全部材料的情况下开放部分馆藏。受访者多次强调信任以及共同实践规范和行为守则的重要性。Jane Winters 以个人档案为例指出，捐赠者、档案工作者和研究者之间的信任关系十分关键，但这种关系既耗时，也具有排他性：如果研究者认识图书馆或档案馆工作人员，就更容易开展这些协商；完全陌生的研究者则困难得多。换言之，访问常常依赖研

究者与档案工作者的一对一沟通。通过谈判并围绕材料使用建立信任关系，原本关闭的馆藏可能被解锁；但依靠个人关系临时开放并不是长期方案。

Providing ad hoc access to known users is not a longstanding solution since it excludes entire groups— for example, independent scholars or family historians who lack a university affiliation and professional networks. As we have seen, access without discrimination is central to GLAM codes of ethics. Adam Nix, Lecturer in Responsible Business at the University of Birmingham, favours a systematic policy of access for those who respect certain codes of behaviour:

只向熟悉的用户临时开放并不能作为长期解决办法，因为这会排除没有大学隶属关系和专业网络的独立学者、家族史研究者等群体。正如前文所述，不歧视地提供访问本就是 GLAM 伦理准则的重要原则。伯明翰大学的 Adam Nix 主张建立系统性访问政策，让愿意遵守明确行为规范的用户获得访问。

> *I would like to see more focus on sharing that duty of care between the archives and the users. I think users need to take a far more active and conscious role in maintaining . . . the integrity of the archival discovery process. And I think that can be done by having well developed and well-respected codes of conduct that the user very consciously agrees with beforehand, which means that even if they see sensitive information, that sensitive information goes no further than that individual researcher.31*
>
> *“我希望更多关注档案机构与用户共同承担这种注意义务。我认为用户需要更主动、更自觉地维护……档案发现过程的完整性。可以通过制定完善且受到尊重的行为守则来实现，让用户事先明确同意这些规范。这样，即使他们看到了敏感信息，这些信息也不会传播到该研究者之外。”31*

This proposed process of opening up archives to users who behave in an ethical manner is based on trust. But it is also based on the possibility of sanctions (including legal sanctions) for those who transgress codes of conduct. A similar system of trust and possible sanction is described in Frances Harris and Fergus Lyon’s research on collaborations across professional cultures. Their central findings are that first, trust is a vital ingredient when collaborating across disciplines and sectors. Second, maintaining collaboration across professional cultures presents particular challenges for building trust. Third, trust in interdisciplinary and transdisciplinary teams is based on norms, information, sanctions, and controls (Harris and Lyon, 2013).

这种向遵守伦理规范的用户开放档案的机制建立在信任之上，同时也依赖对违反行为规范者实施制裁——包括法律制裁——的可能性。Frances Harris 和 Fergus Lyon 关于跨职业文化协作的研究也描述了类似的“信任+制裁”体系：第一，信任是跨学科、跨部门合作的关键；第二，跨越不同职业文化维持合作会给建立信任带来特殊挑战；第三，跨学科和超学科团队中的信任建立在规范、信息、制裁和控制之上。

John Sheridan, Digital Director at The National Archives UK, is in favour of quantifying risk and therefore trust. ‘Looking at our processes around digitization, all tend to lean much more heavily on managing the risks through expert knowledge, than on systems’, Sheridan said.32 He advocates for a computational approach using statistical models to assess risk. To this purpose, TNA has explored a Bayesian network, i.e. a graphical model that represents a set of variables and conditions, often used for probability analysis (Barons et al., 2021). For Sheridan, archival institutions should strike the right balance between risk and access to potentially sensitive materials. It is not a case of ‘transparency above everything’, as:

英国国家档案馆数字平台主管 John Sheridan 主张对风险、因而也对信任进行量化。他指出，目前数字化相关流程更多依赖专家知识管理风险，而不是系统化机制，因此支持使用统计模型进行风险评估。为此，TNA 探索过贝叶斯网络——一种表示变量及条件关系、常用于概率分析的图模型。Sheridan 认为，档案机构需要在风险与潜在敏感材料的开放之间找到正确平衡，而不是把“透明度置于一切之上”。

> *archives are not Wikileaks, and we’re not in the WikiLeaks business. . . . It’s not responsible to data subjects; it’s not responsible to other people’s intellectual property rights; it’s not lawful. So, we then need to build the techniques to provide access responsibly.*
>
> *“档案馆不是 WikiLeaks，我们做的不是 WikiLeaks 的事情……那样做既不对数据主体负责，也不尊重他人的知识产权，而且不合法。因此，我们需要建立能够负责任地提供访问的技术。”*

Until systematic risk models are an inherent part of archives’ digital practice, Sheridan recommends what he calls ‘gradating access’. A process that ‘maximises use but manages the risks of publication’, gradating access can lead to a more

flexible system of publishing potentially sensitive records (but not in breach of data protection laws), by identifying varying levels of risk, and determining necessary exemptions.33

在系统性风险模型尚未成为档案数字实践内在组成部分之前，Sheridan 建议采用他所说的“分级访问”（gradating access）。这种机制旨在“最大化使用，同时管理公开风险”：通过识别不同风险等级并确定必要豁免，可以更灵活地开放潜在敏感记录，同时不违反数据保护法律。

As the number of born-digital records continues to increase, and the capacity of expert assessment becomes overstretched, AI technologies can enable risks to be managed and reduced. Several interviewees told us that whilst AI tools could support the access and use of digital archives, there were still barriers to scaling up experiments and widespread acceptance of its usage. These concerns can be roughly categorised as accountability, control, and bias. We suggest that these concerns could be addressed through closer collaboration between civil servants, GLAM professionals and academics (both in the humanities and computer science).

随着原生数字记录持续增加，专家人工评估能力日益超负荷，AI 技术可以帮助管理和降低风险。多位受访者认为，AI 工具虽然能够支持数字档案的开放和使用，但实验的规模化以及对其广泛接受仍存在障碍。这些担忧大致可归为问责、控制和偏见三类。作者认为，通过公务员、GLAM 专业人士以及人文与计算机科学学者之间更紧密的合作，可以应对这些问题。

3.1 Accountability

3.1 问责

The need to be accountable and transparent about the work the government does is a fundamental part of the codes of practice and professional ethics civil servants work within. Accountability and transparency are also embedded within the Public Records Act. The tools used to aid decisions about the government’s records sent to The National Archives are subject to this need for accountability. James Lappin, Digital Era lead for KIRM34 at the Cabinet Office, explained that if a decision is made to use an AI tool, then ‘the people who deploy it are accountable for the outcomes’ and ‘you’ve got to defend your algorithm . . . the process by which you trained your algorithm, and the process by which you monitored your algorithm’.35

对政府工作承担责任并保持透明，是公务员职业规范和职业伦理的基本组成部分，《公共记录法》也嵌入了问责与透明原则。用于协助判断哪些政府记录应移交国家档案馆的工具，也必须符合这种问责要求。内阁办公室 KIRM34 数字时代负责人 James Lappin 指出，一旦决定使用 AI 工具，“部署它的人就要对结果负责”，并且“必须能够为算法辩护……说明算法是如何训练的，又是如何被监测的”。35

Reticence over the implementation of AI technologies is often connected to the fear of unexpected or unintended results, and where the responsibility for those lies. Clifford Lynch (Coalition for Networked Information) noted that the algorithm often gets blamed, but incorrectly: ‘I don’t get mad at the algorithm, I don’t get mad at the person who wrote the algorithm, I get mad at the people who are applying the algorithm believing that it’s doing the right thing and not understanding what they’ve got there’.36 Lynch’s comments emphasise both the need for those who implement AI tools to understand and be accountable for them, and the common misconception that the algorithm can and should be blamed. Andrew Blick, Head of the Department of Political Economy at King’s College London, reminded us that the controversy surrounding the AI-driven A-Level grade assignment undertaken in 202037 was initially blamed on the algorithm. But public discourse rejected that and demanded accountability for the implementation of the algorithm.38 This requirement for accountability was articulated in other interviews through the need for explainable AI.

人们对实施 AI 技术的犹豫，往往来自对意外或非预期结果以及责任归属的担忧。Clifford Lynch 指出，人们经常错误地责怪算法本身；他不会责怪算法或编写算法的人，而会责怪那些在不了解工具实际情况的前提下就应用算法、相信它会“做对事情”的人。这强调了实施 AI 工具者必须理解工具并对其负责，也揭示了“算法本身应该被归咎”的常见误解。Andrew Blick 以 2020 年英国 A-Level 成绩分配争议为例：最初舆论把责任推给算法，但随后公共讨论转而要求实施算法的人承担责任。其他访谈中，这种问责要求常被表达为对“可解释 AI”的需要。

Explainable AI was discussed by seven of our interviewees as a way of ensuring accountability and transparency in the tools being used by the civil service, as well as for the institutions holding digital archives and the researchers trying to access them. Jane Winters (University of London) was explicit that explainable AI was part of building relationships of trust, and that archives were generally good at being aware of issues like trust and transparency.39 This again is linked the professional codes of ethics and frameworks that GLAM professionals work within. The ICA Code of Ethics specifically references the need to be able to record and justify actions, provide impartial services to users and 'the special trust given to them' (International Council on Archives, 1996). Laura Millar explains how archivists work within the codes of ethics, and that impartiality and transparency are of particular importance when considering ethical archival work (Millar, 2017, p. 95).

七位受访者谈到可解释 AI，认为它能够确保公务员使用的工具、保存数字档案的机构以及试图访问档案的研究者都具备问责性和透明度。Jane Winters 明确把可解释 AI 视为建立信任关系的一部分，并指出档案机构通常对信任和透明问题较为敏感。这同样与 GLAM 专业人士遵守的职业伦理准则有关。ICA《伦理守则》明确要求档案工作者能够记录并说明其行动，为用户提供公正服务，并维护“赋予他们的特殊信任”。Laura Millar 也指出，在伦理档案实践中，公正与透明尤其重要。

A real tension emerges between the professional ethics of archivists and the application of AI where there is no scope for explaining how it works. These forms of unexplainable AI are often referred to as 'black box' due to their opacity. The unexplainable nature of black box AI was also a problem for the historians we interviewed. Adam Nix (University of Birmingham) explained that with black box AI systems 'we have something that we can't fully critique as a researcher and we can't fully understand the results we got or the findings we generate, it becomes an aspect of that process that we can't understand'.40 Nix's point is supported by the requirement for accountability highlighted in the UKROI's Code of Practice for Researchers.41

当 AI 无法解释自身如何运作时，档案工作者的职业伦理与 AI 应用之间就出现明显张力。这种不可解释的 AI 因其不透明性常被称为“黑箱”。受访历史学家同样把黑箱 AI 视为问题。Adam Nix 指出，面对黑箱系统，“作为研究者，我们无法充分批判它，也无法充分理解所得结果或产生的发现；它变成研究过程里我们无法理解的一环。”这一观点也符合 UKRIO《研究人员实践守则》对问责的要求。

Whilst the nature of some AI tools conflicts with key ethical principles, computer scientists and programmers developing the AI technology are working to make tools more transparent and explainable. Graham McDonald (University of Glasgow) explained how although a lot of the tools are currently black boxes, work is being done on 'transparency, explainability and ensuring the fairness of predications'.42 McDonald went on to explain that transparency will be one of the most important factors in giving users of AI systems the confidence that they understand what the system is doing.43

尽管某些 AI 工具的特性与关键伦理原则存在冲突，开发 AI 技术的计算机科学家和程序员正努力提升工具的透明度和可解释性。Graham McDonald 指出，虽然目前很多工具仍是黑箱，但研究正集中于“透明度、可解释性以及确保预测公平”。他进一步表示，透明度会成为用户建立信心、理解系统正在做什么的最重要因素之一。

An example of this can be seen through the Digital Sensitivity Review project being undertaken between the Foreign, Commonwealth and Development Office (FCDO), and the consultancy company SVGC in partnership with the University of Glasgow.44 Andrew Dixon, Managing Director of SVGC, explained that in using AI tools to make decisions about public records to be archives, the public has a right to know how those selections are being made. This is of course derived from the civil service's professional code requirement for accountability and transparency. Dixon explained that this influenced the tools they use in the project: 'we're not using neural networks which are unexplainable. . . we've avoided technologies that create random outcomes'—instead opting to use the kind that are 'predictable and repeatable technologies . . . so you can determine the outcome'.45 Here we see how the principles and professional codes of ethics that govern the civil service, GLAM sector and academic researchers are influencing the technology being used.

英国外交、联邦及发展事务部（FCDO）与咨询公司 SVGC、格拉斯哥大学合作开展的“数字敏感性审查”项目提供了一个例子。SVGC 总经理 Andrew Dixon 指出，当 AI 被用于决定哪些公共记录应被归档时，公众有权知道这些选择是如何作出的，这直接来自公务员职业规范中对问责和透明的要求。因此，项目有意不使用无法解释的神经网络，也避开会产生随机结果的技术，而选择“可预测、可重复”的技术，使结果可以被确定。这里可以看到，公务员、GLAM 领域和学术研究者的职业伦理原则会直接影响实际采用的技术。

McDonald, who also works on this project, pointed out that intuitive explanations of what is happening when you apply an AI tool is crucial for building trust—especially when those using the tool are not the computer scientists who have written the algorithms, tested, and evaluated them.46 Successful AI projects like these are a product of collaborations between the civil servants who manage the sensitivity review process, with archival partners, academic researchers, and computer scientists who provide expertise on the technology. Stakeholders effectively communicated professional principles and codes, and through that, AI tools that are congruent with the notions of accountability and transparency were implemented.

同样参与该项目的 McDonald 指出，直观解释 AI 工具实际在做什么，对于建立信任至关重要，尤其当使用工具的人并不是编写、测试和评估算法的计算机科学家时。这类成功 AI 项目来自多方协作：管理敏感性审查流程的公务员、档案合作伙伴、学术研究者以及提供技术专业知识的计算机科学家共同参与。各利益相关者有效沟通职业原则和规范，从而实施与问责、透明理念相一致的 AI 工具。

3.2 Control

3.2 控制

The second tension that emerged through our interviews focused on the lack of control and trust in AI tools. Eleven of our interviewees brought up the idea of control in discussing the application of AI tools to digital archives. Although we do not want to present colleagues in the civil service, GLAM sector and academia as control obsessed, it should be acknowledged that in these professions, control over what is created, released, archived, described, included, and excluded is important. For civil servants and GLAM professionals, there is a professional responsibility toward maintaining privacy and managing sensitive information. James Lappin (Cabinet Office) explained how once a decision is made to release a document, there is no going back: ‘If you use AI to determine access permissions, to open up access . . . it’s an irrevocable decision, because once you’ve given me that access, there isn’t much point in then taking it away from me, and the horse has bolted, if you like’.47 Similarly, Leontien Talboom (The National Archives UK/University College London) discussed how archivists tend to default to wanting material to only be available in reading rooms where there can be more control over the type of access to it.48 Likewise, Andrew Riley, Senior Archivist at the Churchill Archival Centre, explained how archivists value control and that the loss of that through the use of AI with respect to sensitive data was one of his concerns.49

访谈中出现的第二类张力，集中在对 AI 工具缺乏控制和信任。11 位受访者在谈论 AI 应用于数字档案时提到“控制”。虽然作者并不想把公务员、GLAM 专业人士和学者描绘成“痴迷控制”，但这些职业确实十分重视对哪些内容被创建、发布、归档、描述、纳入或排除的控制。公务员和 GLAM 专业人士还承担维护隐私、管理敏感信息的职业责任。James Lappin 指出，文件一旦开放就无法真正收回：“如果你用 AI 决定访问权限、开放访问……这是不可逆的，因为一旦你让我看到了，再取消访问也没有多少意义，事情已经发生了。”47 Leontien Talboom 也指出，档案工作者往往倾向于只允许在阅览室使用材料，因为那里的访问方式更容易控制。Andrew Riley 同样强调，档案工作者看重控制，对敏感数据使用 AI 可能造成的控制丧失正是他的担忧之一。

These responsibilities towards protecting privacy and sensitive information connects to the code of practice and professional ethics that govern the sectors. The Civil Service’s principle of integrity is particularly relevant here as it references the obligation to undertaking duties responsibly, professionally, and also taking into account the ethical standards that govern other professionals.50 The ICA Code of Ethics is more explicit in the archivist’s responsibilities towards privacy, specifically noting that archivist should ‘take care that corporate and personal privacy as well as national security are protected without destroying information, especially in the case of electronic records where updating and erasure are common practice’ (International Council on Archives, 1996).

保护隐私和敏感信息的责任，与各行业的职业规范和职业伦理直接相关。《公务员守则》中的廉正原则尤其重要，因为它要求公务员负责任、专业地履行职责，并考虑规范其他职业的伦理标准。ICA《伦理守则》对档案工作者的隐私责任规定得更明确，要求档案工作者在不毁坏信息的前提下保护机构和个人隐私以及国家安全，尤其是在电子记录中，因为更新和删除是常见操作。

The fear over a loss of control was articulated differently by the Humanities academics interviewed during the project. In most cases, the lack of control came down to not knowing what was being missed. Historians highlighted the importance of serendipitous findings when accessing a box of physical archival items. Lindsay Aqui, Research Fellow at the University of

Westminster, explained how she felt an ‘immediate discomfort’ around AI-assisted searches because she ‘wouldn’t want there to be something in the process that meant 10% was being excluded because it was deemed irrelevant or didn’t match the search criteria in a way that I thought it would’.51 This sentiment was echoed by Emily Robinson, Senior Lecturer in Politics at the University of Sussex, who noted that using AI-driven search tools may prove a barrier to finding the things you did not know you were trying to find in the archive.52 Helen McCarthy, Professor in Modern and Contemporary British History at the University of Cambridge, described her feelings towards the application of AI to digital archival collections as ambivalent: simultaneously open to the potentials, but nervous about the loss of control for civil servants and archivist, and the potential losses in material that may come of this.53

人文学者对“失去控制”的担忧表达方式有所不同，多数情况下，这种担忧来自不知道自己遗漏了什么。历史学家强调，在翻阅实体档案盒时偶然发现材料的重要性。Lindsay Aqui 表示，AI 辅助检索让她“立即感到不安”，因为她不希望检索过程因为系统认为某些内容无关或不符合预设标准，就排除 10%的材料。Emily Robinson 也认为，AI 驱动检索可能阻碍研究者发现那些事先根本不知道自己要寻找的东西。Helen McCarthy 则把自己对 AI 数字档案应用的态度描述为矛盾：一方面愿意接受其潜力，另一方面又担心公务员和档案工作者失去控制，以及因此可能损失材料。

One way in which computer scientists are attempting to address this concern over a loss of control is by flipping the narrative on the use of AI. The term ‘human-in-the-loop’ is often used to refer to the way in which AI tools are trained with human interaction. Graham McDonald explained how ‘if you’re putting across a narrative that it’s human in the loop, there’s an expectation that the Artificial Intelligence is going to be the one that’s making the decisions and doing the job. But that’s not right—that’s not actually what it is expected to be in practice’.54 Instead, McDonald argued that we should focus more on ‘computer in the loop processes where it’s the AI that is there to help and assist the human in the decision-making process’.55 He suggested that ‘these little changes in narrative can help to change the perspectives of how people actually see it’.56 Shifting perspectives in this way may offer some alleviation to the fears that as responsible civil servants, archivists, or researchers might no longer have control over their own work remit.

计算机科学家应对“失去控制”担忧的一种方法，是改变描述 AI 的叙事。“human-in-the-loop（人在回路中）”常用于描述 AI 工具如何通过人类互动进行训练，但 McDonald 指出，这种表述容易让人以为 AI 才是作出决定、完成工作的主体，而实际并非如此。他建议更强调“computer in the loop（计算机在回路中）”：AI 的角色是帮助并辅助人类作出决定。这样的细微叙事变化，可以改变人们理解 AI 的方式，并缓解公务员、档案工作者或研究者担心失去对自身工作控制的焦虑。

3.3 Bias

3.3 偏见

The third tension that emerged in our interviews was that of bias. Twelve of our interviewees highlighted that algorithmic bias was a concern for them or others around them. The notion of bias is particularly pertinent to the codes of practice and professional ethics across the civil service, GLAM sector, and academia. Impartiality is a key feature of the Civil Service code and specifically noted that a civil servant must not ‘act in a way that unjustifiably favours or discriminates against particular individuals or interests’.57 This sentiment is echoed in the ICA Code of Ethics which states that archivists should ‘provide an impartial service’ (International Council on Archives, 1996, p. 2) and UKRIO Code of Practice emphasizes the importance of recognizing the dignity, rights, and safety of people in research.58

访谈中出现的第三类张力是偏见。12 位受访者表示，算法偏见是他们或周围人的担忧。偏见问题与公务员、GLAM 领域和学术界的职业规范尤其相关。《公务员守则》把公正列为核心要求，明确规定公务员不得以缺乏正当理由的方式偏袒或歧视某些个人或利益；ICA《伦理守则》同样要求档案工作者“提供公正服务”，UKRIO《研究人员实践守则》也强调研究中应尊重人的尊严、权利和安全。

Oonagh Murphy, Lecturer in Arts Management at Goldsmiths University, described how in working with AI and the GLAM sector there is scope for what she framed as bias squared. Murphy explained how many galleries, libraries, museums, and archives already have biases built into their collections through hundreds of years of colonialism, and the forms and structures that emerged from that and influenced how collections were put together and described. The use of AI in institutions with such collections risked adding additional elements of bias—compounding the issue.59 The danger of this

happening was recognized not just by those who work in or research about the GLAM sector, but also by Se´bastien A. Krier (AI Policy and Governance Advisor) who acknowledged that training AI tools on cultural collections would encounter issues such as sexism.60 Likewise, historians like Lindsay Aqui explained that they had ‘a bit of suspicion around what might end up excluded from search results . . . because archives are already curated and people’s voices, stories, experiences do get excluded from the archive, so there is a risk that some AI tools may further marginalise already marginalised people’.61 The potential to reinforce bias is a genuine concern for many stakeholders.

Goldsmiths University 艺术管理讲师 Oonagh Murphy 把 AI 与 GLAM 领域结合时可能出现的问题称为“偏见的平方”。她指出，许多画廊、图书馆、博物馆和档案馆本身的馆藏，已经在数百年殖民主义及其形成的制度结构中积累了偏见，这些结构影响了馆藏如何被建立和描述。在这样的机构中使用 AI，可能进一步叠加新的偏见。GLAM 从业者之外的 AI 政策人士也意识到这一危险，例如 Sébastien A. Krier 承认，在文化馆藏上训练 AI 会遇到性别歧视等问题。Lindsay Aqui 也担心，档案本身已经经过筛选，一些人的声音、故事和经验早已被排除，AI 工具可能进一步边缘化本就被边缘化的群体。

Whilst bias should not be embedded as normative practice, several interviewees recognized that bias can never be fully removed. Jane Winters noted how Humanities researchers tend to understand that bias is always present, and that acknowledging bias and understanding what it means for your research is important: ‘you can’t get rid of biases, you just have to acknowledge that they’re there, and understand what that means for the results that you get. . . if you try and get rid of bias, you just introduce a new one, so it’s acknowledging that it’s there and working with that, rather than trying to get rid of it’.62 Gareth Millward, Research Fellow at the University of Birmingham, was also keen to note that researchers need to be ‘hyperaware of the biases that can be baked into some of the technologies that we use . . . and at least be able to contextualise them if not be able to eliminate those biases altogether’.63

虽然偏见不应成为规范性做法，多位受访者也认识到，偏见不可能被彻底消除。Jane Winters 指出，人文学者往往理解偏见始终存在，重要的是承认偏见并理解它对研究结果意味着什么：“你无法消除偏见，只能承认它们存在，并理解它们对结果意味着什么……如果试图消除一种偏见，你往往只是引入另一种。因此应该承认它的存在并与之共处，而不是试图把它完全去除。”Gareth Millward 也强调，研究者需要高度意识到技术中可能内置的偏见，即使不能消除，也至少要能够把它们放回具体语境中解释。

4 Collaborations and building trust

4 协作与建立信任

This section now seeks to explore how the deeper collaborations between civil servants, GLAM sector professionals, and both humanities and computer science academics can help establish appropriate levels of trust in AI tools and support their use in making digital archives more accessible and usable. As professions, the civil service, GLAM sector and academia all value collaborative working. Within the Civil Service Code, there are references to co-operation and respect with colleagues and partners, as well as to taking into account the ethical standards governing other professions.64 The ICA Code of Ethics states that archivists ‘should cooperate with members of related professions on the basis of mutual respect and understanding’ (International Council on Archives, 1996, p. 3), and the UKRIO Code of Practice advises that researchers working collaboratively do so with awareness of the ethical guidelines and standards of collaborators.65 In practice, however, civil servants, GLAM professionals and academics seldom collaborate to unlock digital archives.

本节考察公务员、GLAM 专业人士以及人文和计算机科学学者之间更深入的合作，如何帮助建立对 AI 工具的适当信任，并支持其用于提高数字档案的可获取性和可使用性。作为职业群体，公务员体系、GLAM 领域和学术界都重视合作。《公务员守则》要求与同事和合作伙伴合作、相互尊重，并考虑其他职业所遵循的伦理标准；ICA《伦理守则》要求档案工作者“在相互尊重和理解的基础上，与相关职业成员合作”；UKRIO《研究人员实践守则》也要求合作研究者了解协作者的伦理指南和标准。但在实践中，这些群体很少真正协作来解锁数字档案。

Wider, cross-disciplinary discussions are required: discussions about shared concerns, shared professional and ethical principles, as well as solutions with the computer scientists developing the tools, rather than a focus purely on the barriers to implementing AI. The three main tensions discussed in the previous section contribute to a general mistrust of AI throughout the professionals working in the civil service, GLAM sector and academia, but within each of the tensions, there

were interviewees who identified potential solutions that were compliant with the different codes of practice and professional ethics.

因此需要更广泛的跨学科讨论：不仅讨论共同的担忧、共同的职业和伦理原则，还要与开发工具的计算机科学家共同讨论解决办法，而不能只关注实施 AI 的障碍。上一节讨论的问责、控制和偏见三类张力，使公务员、GLAM 专业人士和学术界普遍对 AI 缺乏信任；但在每一种张力中，也有受访者提出了符合不同职业规范和职业伦理的潜在解决方案。

A central takeaway is that interdisciplinary partnerships are vitally important in developing trustable AI tools. John Sheridan (The National Archives UK) noted:

一个核心结论是：跨学科伙伴关系对于开发值得信任的 AI 工具至关重要。英国国家档案馆的 John Sheridan 指出：

> *It's all about the knowledge transfer into our sector from outside. No one is building anything specifically for us. It is all about things that are being built in the world, and then someone's going, 'Oh, I could apply this to archives.' And I think all we can really do is equip ourselves with learning to understand what that kind of technology transfer process looks like and understand what the implications are.*
>
> *"关键在于把外部知识转移进我们的领域。没有人专门为我们开发这些东西，都是世界上先出现某种技术，然后有人说，'哦，这个也许可以用于档案。'我们真正能做的是通过学习，让自己理解这种技术转移过程是什么样的，以及它会带来什么影响。"*

Sheridan went on to explain how 'it's quite hard for us to know how to navigate this world where, whilst we have collections, we're effectively, certainly in terms of budgets, and then in terms of capability, we're fleas around elephants. But we're not on our own. These issues are pervasive, and again, I think it's about not just talking to each other, but talking to other people who share concerns, is going to be very important, too'.66

Sheridan 进一步说，档案领域很难知道如何在这个世界中行动：它虽然拥有馆藏，但无论预算还是技术能力，相对于大型技术力量都像"大象旁边的跳蚤"。不过，档案机构并不孤单，这些问题广泛存在。因此，重要的不只是行业内部彼此交流，也要同其他拥有相似关切的人对话。

Daniel van Strien, Digital Curator at the British Library also emphasized that the GLAM sector needs to be actively involved in how AI tools are applied to collections. If these tools are developed without engaging with GLAM professionals, they may end up not being useful for the sector in practice.67

大英图书馆数字馆员 Daniel van Strien 同样强调，GLAM 领域必须主动参与 AI 工具如何被应用于馆藏。如果工具在开发时没有与 GLAM 专业人士沟通，最终可能根本无法满足该领域的实际需求。

Whilst some interviewees expressed mistrust over the idea of AI as a tool, there were also expressions of trust in the archive's decisions around potential uses of AI. Gareth Millward (University of Birmingham) explained how 'if The National Archive is providing me with the tool to search its own archive, I trust that they have thought about that, and that is part of their archival process. . . just as I have to put my faith in their cataloguing technique'.68

虽然一些受访者对 AI 作为工具持不信任态度，但也有人信任档案机构关于 AI 潜在用途的决定。Gareth Millward 说，如果英国国家档案馆向他提供搜索自己馆藏的工具，他会相信该馆已经对此进行过认真考虑，并把它作为档案流程的一部分，就像研究者也必须信任其编目方法一样。

In practice, however, not many users have the knowledge and ability to use advanced technologies such as AI to explore archives. In turn, this skills gap hinders collaborative discussions. One professional in a large archival institution in London told us:

现实中，能够使用 AI 等先进技术探索档案的用户并不多，这种技能缺口反过来妨碍协作讨论。一位伦敦大型档案机构的专业人士告诉研究者：

> *we have had a few attempts at trying to run events for researchers to find out what they might be expecting, or what they might want. And . . . the majority of our existing core research audience is not yet working with born-digital material.*

“我们曾尝试举办活动，了解研究者可能期待什么、想要什么。但……我们现有的核心研究受众大多数还没有在使用原生数字材料。”

This is particularly frustrating for archivists and digital preservation specialists who have been working on these kinds of records for years. ‘The events turn into a training event on what the archive profession has learnt from 10 years of trying to do digital preservation, rather than really extracting expectations from researchers’. For this professional, it was surprising to see little demand from researchers wanting to use digital tools to explore a digital archive. ‘A bit like the depositors, [researchers] transfer their paper world and their paper research methods to the digital world’.69

这对已经多年处理此类记录的档案工作者和数字保存专家而言尤其令人沮丧。活动往往变成档案行业向研究者培训“过去十年我们在数字保存方面学到了什么”，而不是从研究者那里真正了解其需求。该受访者惊讶地发现，很少有研究者主动要求使用数字工具探索数字档案；“有点像捐赠者一样，[研究者]把他们的纸本世界和纸本研究方法直接搬到了数字世界”。

Andrew Riley (Churchill Archival Centre) also mentioned the lack of skills which prevents users from productively working in digital collections. He mentioned his experience of attending a workshop at the Foreign Office organised by historian Helen McCarthy on born-digital records. Riley discovered that like many ‘traditional’ archivists trained to process paper records, historians at the workshop felt insecure when it came to using born-digital materials:

Churchill Archives Centre 的 Andrew Riley 也提到，技能不足阻碍用户有效使用数字馆藏。他参加过由历史学家 Helen McCarthy 在外交部组织的原生数字记录工作坊，发现很多历史学家和接受纸质记录训练的“传统”档案工作者一样，在面对原生数字材料时缺乏信心：

I thought they’d be much more self-confident, but they all seemed to doubt their essential skill sets as historians in using this kind of material and, for similar reasons, hoping to be retired too before they had to do too much of this, which really surprised me.70

“我原以为他们会自信得多，但他们似乎都怀疑自己作为历史学家的基本技能能否应对这种材料；出于类似原因，他们甚至希望自己在不得不大量处理这些材料之前就已经退休，这让我很惊讶。”70

In this context, it is not surprising that many researchers turn to more easily accessible and usable collections: paper collections but also digitized collections searchable via keyword search. Concerns about big data and Artificial Intelligence could be allayed through a conceptual understanding of the techniques used. Yet, humanities scholars have few institutional opportunities to fill the gaps in their knowledge around AI. For example, the 2022 ‘Digital Humanities at Oxford’ Summer School did not include any course on AI.

在这种情况下，许多研究者转向更容易获取和使用的馆藏并不令人意外，包括纸质馆藏以及可以通过关键词检索的数字化馆藏。若能从概念层面理解大数据和 AI 所使用的技术，相关担忧可能得到缓解，但人文学者在机构层面很少有机会弥补 AI 知识缺口。例如，2022 年牛津“数字人文”暑期学校就没有开设任何 AI 课程。

Without significant pressure from researchers to open up born-digital collections, and without any easy solutions to the issues of sensitivity and copyright, it can seem rational to keep archives closed for now. James Lappin thus praised NARA’s decision to preserve the Whitehouse digital archive, even if it is still “dark.” Preservation was presented as the most important thing, since it makes possible access at a later stage.71 Breanda´n Knowlton (who works in a government team) pushed for openness and told us:

如果研究者没有强烈要求开放原生数字馆藏，而敏感性和版权问题又没有简单解决方案，那么暂时保持档案关闭似乎很合理。James Lappin 因此赞扬 NARA 保存白宫数字档案的决定，即使这些档案仍处于“黑暗”状态；保存被视为最重要的，因为它至少让未来开放成为可能。不过，政府团队成员 Breandán Knowlton 更主张开放，他说：

When you’re talking about the public sector, the work we do is funded by public budgets or scrutinised by Parliament and by journalists and by everybody else. Most of our work should be public by default. . . I think if more of our work was public, I think more people would find it, kind of, boring. But also, it wouldn’t be [in case of data leaks] ‘Ooh, secret trove of Cabinet Office emails released by aide’.72

“谈到公共部门，我们的工作由公共预算资助，也受到议会、记者以及其他人的审视。我们绝大多数工作原则上应该默认公开……如果更多工作公开，我想人们反而会觉得它很无聊。但这样一来，[如果发生数据泄露]也不会变成‘哇，内阁办公室一批秘密邮件被助手泄露了’。”72

As we have seen, accountability and transparency are central to civil servants’ codes of ethics, but this should be balanced against the need to protect national security and avoid leaking sensitive records. Having more collaborative discussions with other stakeholders (including archivists and academics) would offer government professionals new perspectives and allow them to make informed decisions on records to transfer to archival repositories.

正如前文所示，问责与透明是公务员伦理准则的核心，但它们必须与保护国家安全、避免敏感记录泄露的需要相平衡。若政府工作人员能与其他利益相关者——包括档案工作者和学者——开展更多协作讨论，就能获得新的视角，并对哪些记录应转交档案机构作出更有依据的判断。

5 Conclusion

5 结论

This article has shown that the lack of communication between civil servants, GLAM professionals and academics often leads to mistrust of other stakeholders. Interviewees anticipated situations where data would leak, reputations would be damaged, and in the worst cases, national security would be at risk. But there can be no betrayal without trust. And it is precisely the lack of trust in other stakeholders, and in new technologies, that was highlighted in our interviews. Mistrust of technology makes it difficult to implement AI tools. The concerns highlighted by interviewees—in terms of accountability, control, and bias—make it difficult to accelerate the development of AI applied to archives. The article recommends more cross-sector and cross-disciplinary dialogue to surface the shared professional ethics between stakeholders. Deeper understanding between humans is an essential step to apply AI to born-digital records. More research is also needed across disciplinary boundaries. Indeed, the issues of innovative technologies applied to digital archives have too often been explored within single disciplines such as archival studies or computer science. Yet, applying AI to archives requires a variety of skillsets: not only computational skills, but also domain knowledge, i.e. knowledge about the entities that are being treated by the algorithms. Thinking within disciplinary boxes hinders our collective ability to apply technology to digital records. Digital Humanities—which is in essence an interdisciplinary field—has an important role to play in bridging the gap between record creators, archivists, researchers, and other users.

本文表明，公务员、GLAM 专业人士和学者之间缺乏沟通，经常导致对其他利益相关者的不信任。受访者担心数据泄露、声誉受损，最严重时甚至可能危及国家安全。但没有信任，就谈不上背叛。访谈中真正突出的问题，正是对其他利益相关者以及新技术缺乏信任。对技术的不信任使 AI 工具难以实施；围绕问责、控制和偏见的担忧，也减缓了 AI 在档案领域的发展。本文建议加强跨行业、跨学科对话，把不同利益相关者共享的职业伦理明确呈现出来。人与人之间更深入的理解，是把 AI 应用于原生数字记录的必要前提。与此同时，还需要更多跨学科研究。创新技术与数字档案的问题过去过多被放在档案学或计算机科学等单一学科内部讨论，但 AI 应用于档案需要多种技能，不仅包括计算能力，也包括领域知识——即理解算法正在处理的对象。固守学科边界会削弱我们共同把技术应用于数字记录的能力。数字人文本质上是跨学科领域，因此在连接记录创建者、档案工作者、研究者和其他用户方面具有重要作用。

Notes / 注释

1. Born-digital archives refer to archival materials that originated in a digital format (for examples: emails, PDFs, Word documents, audio, and video digital files). These born-digital records differ from “digitised” materials created through processes such as scanning and photographing.

1. 原生数字档案是指从一开始就以数字格式产生的档案材料，例如电子邮件、PDF、Word 文档、音频和数字视频文件。它们不同于通过扫描、摄影等过程从实体材料转化而来的“数字化”材料。

2. Artificial intelligence (AI) refers to the use of computational processes to learn, make decision, and solve problems. Machine learning (ML), which is often referenced in discussions about AI, is an application of it. ML is the process by which a computer system is able to continue learning and improving on its own based on previous processes it has undertaken. ML is considered a subset of AI.

2. 人工智能（AI）指利用计算过程进行学习、作出决策和解决问题。讨论 AI 时经常提到的机器学习（ML）是 AI 的一种应用形式：计算机系统能够根据此前执行的过程继续自行学习和改进。机器学习通常被视为 AI 的子领域。

3. For example, AURA (Archives in the UK/ Republic of Ireland and AI) www.aura-network.net; AEOLIAN (Artificial Intelligence for Cultural Organisations) www.aeolian-network.net; and AI4LAM https://sites.google.com/view/ai4lam. It should be noted that AI is also enthusiastically embraced in relation to digitized collections (not only born-digital collections).

3. 例如 AURA（英国/爱尔兰共和国档案与 AI）、AEOLIAN（文化机构人工智能）以及 AI4LAM。需要注意的是，AI 也被积极用于数字化馆藏，而不仅仅是原生数字馆藏。

4. The National Archives has a lifespan assumption that governs their approach to personal data when the subject is not personally known to the archive. It states that: “Given the large number of individuals commonly featuring in archive collections, archive services will not be in a position to ascertain whether they are still alive. If it is not known whether a data subject is alive or dead, the following working assumptions can be used: Assume a lifespan of 100 years; If the age of an adult data subject is not known, assume that they were 16 at the time of the records; If the age of a child data subject is not known, assume person was less than 1 at the time of the records. If the individual is known to be more than 100 years old and still living then compliance with data protection law is still required.” (The National Archives, 2018, p. 34).

4. 英国国家档案馆在无法确认数据主体实际生死情况时，以预设寿命来处理个人数据。其工作假设包括：寿命按 100 年计算；如果成年数据主体年龄不明，则假定记录形成时其年龄为 16 岁；如果儿童数据主体年龄不明，则假定记录形成时其年龄不足 1 岁。若已知某人超过 100 岁且仍在世，仍须遵守数据保护法律。（The National Archives, 2018, p. 34）

13. Interview with James Baker, 22 June 2022.

13. 采访 James Baker，22 June 2022。

14. Interview with Clifford Lynch, 17 June 2021.

14. 采访 Clifford Lynch，17 June 2021。

15. Interview with Jason Baron, 22 June 2021.

15. 采访 Jason Baron，22 June 2021。

17. Interview with Leslie Johnston, 18 June 2021.

17. 采访 Leslie Johnston，18 June 2021。

18. Interview with Jason Baron, 22 June 2021.

18. 采访 Jason Baron，22 June 2021。

19. Ibid.

19. 同上。

21. Interview with Leontien Talboom, 22 June 2021.

21. 采访 Leontien Talboom，22 June 2021。

22. Interview with Jason Baron, 22 June 2021.

22. 采访 Jason Baron，22 June 2021。

24. Ibid.

24. 同上。

25. Interview with Graham McDonald, 23 June 2021.

25. 采访 Graham McDonald，23 June 2021。

27. Interview with Jason Baron, 22 June 2021.

27. 采访 Jason Baron，22 June 2021。

29. Interview with Jane Winters, 24 June 2021.

29. 采访 Jane Winters，24 June 2021。

30. Interview with Jane Winters, 24 June 2021.
30. 采访 Jane Winters，24 June 2021。

31. Interview with Adam Nix, 22 June 2021.
31. 采访 Adam Nix，22 June 2021。

32. Interview with John Sheridan, 29 June 2021.
32. 采访 John Sheridan，29 June 2021。

33. Ibid.
33. 同上。

34. KIM refers to Knowledge, Information and Records Management.
34. KIM 指 Knowledge, Information and Records Management（知识、信息与记录管理）。

35. Interview with James Lappin, 29 June 2021.
35. 采访 James Lappin，29 June 2021。

36. Interview with Clifford Lynch, 17 June 2021.
36. 采访 Clifford Lynch，17 June 2021。

38. Interview with Andrew Blick, 16 July 2021.
38. 采访 Andrew Blick，16 July 2021。

39. Interview with Jane Winters, 24 June 2021.
39. 采访 Jane Winters，24 June 2021。

40. Interview with Adam Nix, 22 June 2021.
40. 采访 Adam Nix，22 June 2021。

42. Interview with Graham McDonald, 23 June 2021.
42. 采访 Graham McDonald，23 June 2021。

43. Ibid.
43. 同上。

45. Interview with Andrew Dixon, 1 July 2021. As explained earlier in this article, we are using a broad definition of AI as a set of computational approaches (rather than focusing on AI in relation to deep learning and neural networks).
45. 采访 Andrew Dixon，2021 年 7 月 1 日。正如本文前面所说明的，我们对 AI 采用宽泛定义，把它理解为一组计算方法，而不是只把 AI 限定为深度学习和神经网络。

46. Interview with Graham McDonald, 23 June 2021.
46. 采访 Graham McDonald，23 June 2021。

47. Interview with James Lappin, 29 June 2021.
47. 采访 James Lappin，29 June 2021。

48. Interview with Leontien Talboom, 22 June 2021.
48. 采访 Leontien Talboom，22 June 2021。

49. Interview with Andrew Riley and Christopher Knowles, 7 July 2021.
49. 采访 Andrew Riley and Christopher Knowles，7 July 2021。

51. Interview with Lindsay Aqui, 1 July 2021.
51. 采访 Lindsay Aqui，1 July 2021。

52. Interview with Emily Robinson, 15 July 2021.
52. 采访 Emily Robinson，15 July 2021。

53. Interview with Helen McCarthy, 1 July 2021.
53. 采访 Helen McCarthy，1 July 2021。

54. Interview with Graham McDonald, 23 June 2021.
54. 采访 Graham McDonald，23 June 2021。

55. Ibid.

55. 同上。

56. Ibid.

56. 同上。

59. Interview with Oonagh Murphy, 30 June 2021.

59. 采访 Oonagh Murphy，30 June 2021。

60. Interview with Sébastien A. Krier, 2 July 2021.

60. 采访 Sébastien A. Krier，2 July 2021。

61. Interview with Lindsay Aqui, 1 July 2021.

61. 采访 Lindsay Aqui，1 July 2021。

62. Interview with Jane Winters, 24 June 2021.

62. 采访 Jane Winters，24 June 2021。

63. Interview with Gareth Millward, 7 July 2021.

63. 采访 Gareth Millward，7 July 2021。

66. Interview with John Sheridan, 29 June 2021.

66. 采访 John Sheridan，29 June 2021。

67. Interview with Daniel van Strien, 2 July 2021.

67. 采访 Daniel van Strien，2 July 2021。

68. Interview with Gareth Millward, 7 July 2021.

68. 采访 Gareth Millward，7 July 2021。

69. Interview with [name withheld], 27 May 2021.

69. 采访 [name withheld]，27 May 2021。

70. Interview with Andrew Riley and Christopher Knowles, 7 July 2021.

70. 采访 Andrew Riley and Christopher Knowles，7 July 2021。

71. Interview with James Lappin, 29 June 2021.

71. 采访 James Lappin，29 June 2021。

72. Interview with Breandán Knowlton, 1 July 2021.

72. 采访 Breandán Knowlton，1 July 2021。

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